EMNA MESSAOUDI



Software Engineering Student — Full-Stack & AI

messaoudiemna75@gmail.com | +216 94 850 334 | Ariana, Tunisia
LinkedIn: linkedin.com/in/messaoudiemna | GitHub: github.com/emnamess | Portfolio: emnamess.github.io

Final-year software engineering student seeking a 6-month end-of-studies internship, available from December 2026 (visa permitting). Full-stack developer (React, Angular, Spring Boot, FastAPI, .NET) with hands-on experience in generative AI (LangChain, RAG, agents) and DevOps (Docker, CI/CD).

PROFESSIONAL EXPERIENCE

Web Developer — Bulgin June 2026 – August 2026

- Developed an HR platform (FastAPI, React) integrated with ZKBioTime to manage attendance, leave and statistics.
- Designed an employee management application with JWT authentication, CRUD operations and data export, connected to a multi-site ZKTeco/ZKBio database.
- Developed a visitor management and access control module with customizable reports and an interactive analytics dashboard.

Technologies: React/Next.js, TypeScript, FastAPI, Python, PostgreSQL, JWT, Vite, GitHub

Web Developer — Nissaf Lighting June 2025 – August 2025

AionTime — AI-Powered Project Management Assistant

- Developed a platform centralizing project management data through the ClickUp, Trello, Asana and Azure DevOps APIs.
- Integrated AI features for task organization and automated generation of progress reports and charts.
- Contributed to the design of an architecture combining frontend, backend and AI services.

Technologies: Next.js, C#/.NET (ASP.NET Core), Python, LangChain, Qdrant, pgvector, Azure DevOps, GitHub

Web & Mobile Developer — Deaf Square February 2024 – June 2024

- Designed and developed an accessible e-learning website and mobile app for deaf people, featuring sign language videos and an interface that supports self-paced learning.
- Followed an Agile Scrum methodology; modeled requirements and features with UML.

Technologies: Angular, Spring Boot, MongoDB, React Native, Bootstrap

ACADEMIC PROJECTS

SkillForge — E-Learning Platform 2026 – Present

- E-learning platform (Django) with four roles, AI-corrected quizzes, live mentoring and certificates; team of 5.

Technologies: Django, Python, UML

Inspecteur IA — Multi-Agent Detective Game · github.com/emnamess/Inspecteur_IA October 2026

- MCP server (7 tools), LangGraph agents (suspects + hallucination-checking verifier), RAG memory (Qdrant), 46 tests, CI.

Technologies: Python, MCP, LangGraph, Qdrant, FastAPI, React, TypeScript, Tailwind, Docker, pytest, GitHub Actions

Smart-Property — Intelligent Real Estate Platform February 2026 – May 2026

- Real estate platform featuring price prediction, property analysis and personalized recommendations.
- Set up a CI/CD pipeline with Jenkins and Docker automating build, testing and deployment.

Technologies: React, NestJS, WebSockets, Docker, Jenkins, CI/CD, Scrum, UML

EDUCATION

Engineering Degree in Computer Science
ESPRIT, Private Higher School of Engineering and Technology | Since September 2024

Bachelor's Degree in IT, Information Systems Development
ISET Radès, Higher Institute of Technological Studies | September 2020 – June 2024

TECHNICAL SKILLS

Programming languages: Python, Java, C#, JavaScript, TypeScript, PHP, C++, SQL
Frontend & UX/UI: React, Next.js, Angular, React Native, HTML, CSS, Tailwind CSS, Bootstrap, UX/UI (Figma)
Backend: FastAPI, Spring Boot, Node.js, NestJS, Symfony, .NET (ASP.NET Core, Entity Framework), Laravel, Django
AI & data: LangChain, RAG, LLM integration, Qdrant, pgvector, PostgreSQL, MySQL, SQL Server, MongoDB (Atlas), Oracle
DevOps & tools: Docker, Kubernetes, Jenkins, CI/CD, Git, GitHub, GitLab, Maven, AWS, Linux, VMware, RabbitMQ and Kafka (basics)
Software design & testing: OOP, MVC, UML, Scrum, Jira, unit testing (JUnit), SonarQube, IDEs: VS Code, IntelliJ IDEA, Eclipse

LANGUAGES

French: fluent (schooling and higher education in French) | English: fluent